

Design and Implement Your First MCP Resource Schema

You've learned the theoretical foundations of MCP and seen real-world implementations. Now it's time to bridge that gap by designing your own MCP resource schema—the foundation of any successful MCP implementation.

Objective

The goal isn't just to list sales data types—it's to design an MCP resource schema that enables dynamic, AI-driven access to customer, deal, and activity data. Are you creating intuitive, scalable URIs and clear resource definitions? Are you documenting access patterns that balance simplicity with functionality, ensuring your AI assistant can deliver accurate, real-time insights across evolving sales needs?

Scenario

Scenario Setup Grounded Narrative



You're working as a senior developer at TechFlow Analytics, a company that provides AI-powered business intelligence tools. Your team is building an AI assistant that helps sales teams analyze customer data, track deal progress, and generate insights.

Currently, your AI assistant can only work with static data exports. But your sales team needs real-time access to customer information, deal statuses, and sales pipeline data. They want to ask questions like "Show me all high-value deals closing this month" or "What's the status of the Johnson account?"

Your task is to design an MCP resource schema that will enable your AI assistant to access this sales data dynamically. The sales team uses a CRM system that tracks customers, deals, and sales activities. You need to design resources that provide meaningful, filtered access to this data.

Instructions for learners

Design a comprehensive MCP resource schema for a sales intelligence AI assistant that needs access to customer data, deal information, and sales activities.

Refine the learner's task with completion criteria

Your MCP resource schema should include

1. Resource Identification

Identify the key data types your AI assistant needs to access:

- List at least 4 different types of sales data (e.g., customers, deals, activities, pipeline stages)
- Consider what information would be most valuable for AI-powered sales insights
- Think about how sales teams actually use this data in practice

2. Resource URI Design

Design intuitive, scalable URI patterns for your resources:

- Create resource URIs that follow consistent naming conventions
- Consider filtering patterns (e.g., "deals://active", "customers://high-value")
- Design URIs that could scale as your system grows

3. Resource Schema Definition

Define the structure and metadata for each resource:

- Specify the data fields each resource will provide
- Define appropriate MIME types and formats
- Include descriptive names and descriptions for each resource

4. Access Pattern Documentation

Document how your AI assistant will use these resources:

- Describe common query patterns the AI might use
- Explain the relationship between different resources
- Consider how resources might be combined for complex insights

Tools to Use

- - Text editor or note-taking app
- - JSON formatting tool (optional)
- - Your favorite documentation format

Submission

Deliverable Format

Create a structured document that includes:

- Resource inventory with descriptions
- URI patterns and examples
- Schema definitions for each resource
- Usage documentation

Reflection

Reflection Guidelines

After completing your schema design, reflect on these questions:

- How did you balance simplicity with functionality in your URI design?
- What assumptions did you make about how the AI assistant would use the data?
- How would your schema handle edge cases or unusual data patterns?

- What would you change if you had to support multiple sales teams with different needs?

Engage in reflection Transition to Dialogue



Once you've completed your schema design, you'll have the opportunity to discuss your approach with others and get feedback on your design decisions. This collaborative reflection will help you see alternative approaches and refine your MCP design skills.